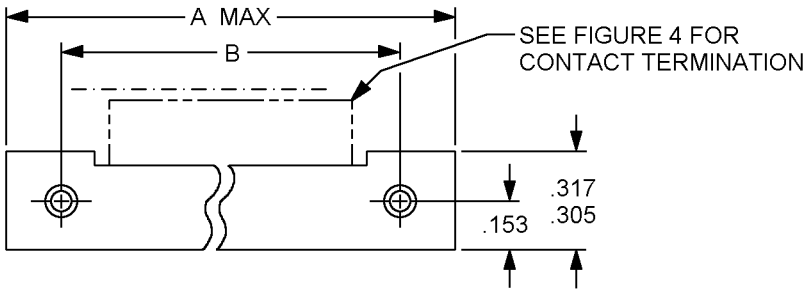
INCH-POUND
MIL-DTL-55302/142C
7 March 2005
SUPERSEDING
MIL-C-55302/142B
17 November 1988

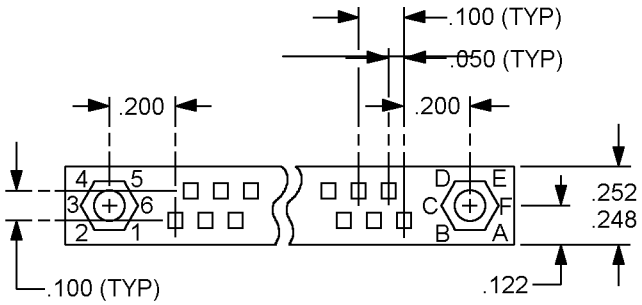
DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, RECEPTACLE WITH SOCKET CONTACTS, FOR PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.





Inches	mm	Inches	mm
.050	1.27	.248	6.30
.100	2.54	.252	6.40
.122	3.10	.305	7.45
.153	3.89	.317	8.05
.200	5.08		

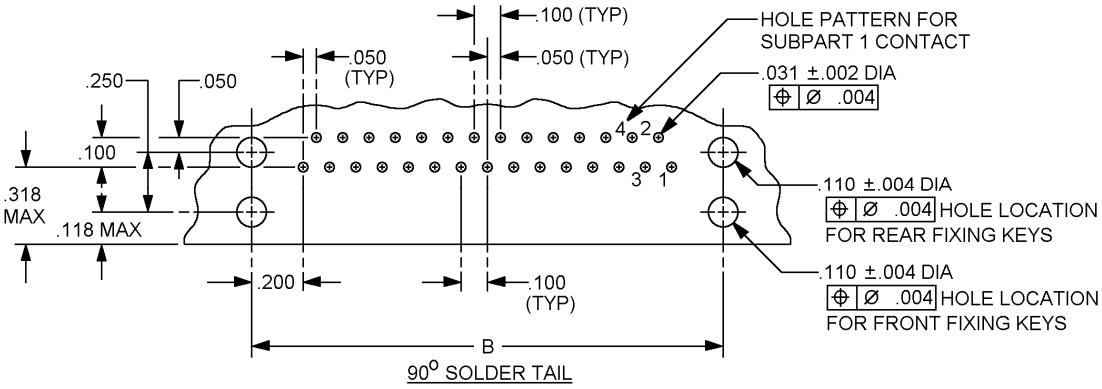
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).

FIGURE 1. Connector, receptacle (.100 inch spacing).

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Arrangements and characteristics.

Contact arrangement	Number of contacts	Dimensions (inches)	
		A max	B ± .006 (0.15)
M55302/142A	17	1.516 (38.50)	1.200 (30.48)
M55302/142B	29	2.114 (53.70)	1.800 (45.72)
M55302/142C	33	2.314 (58.80)	2.000 (50.80)
M55302/142D	41	2.716 (69.00)	2.400 (60.96)
M55302/142E	53	3.315 (84.20)	3.000 (76.20)
M55302/142F	65	3.917 (99.50)	3.60 (91.44)



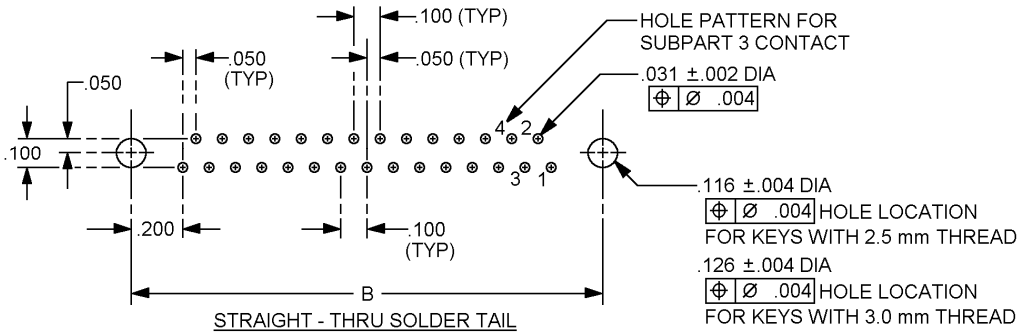
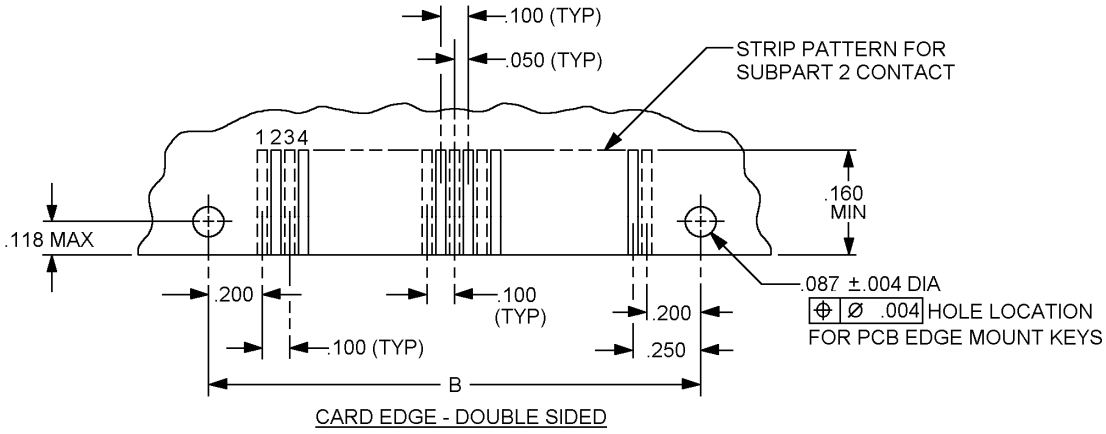


FIGURE 2. Recommended hole patterns.

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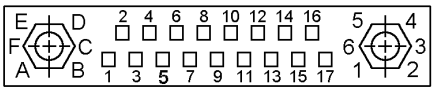
Inches	mm	Inches	mm	Inches	mm
.002	0.05	.100	2.54	.200	5.08
.004	0.10	.110	2.79	.250	6.35
.025	0.64	.116	2.95	.318	8.08
.031	0.79	.118	3.00		
.050	1.27	.126	3.20		
.087	2.21	.160	4.06		



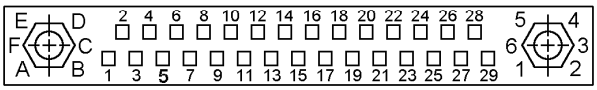
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).
4. Metric equivalents are in parentheses.

FIGURE 2. Recommended hole patterns – Continued.

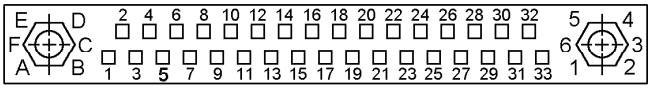
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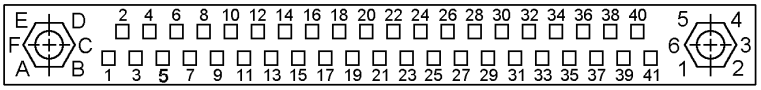
PART NUMBER
M55302/142A



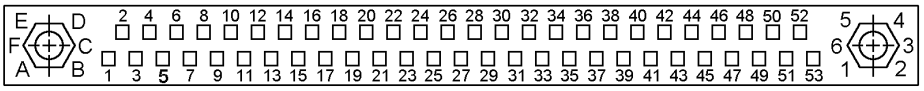
PART NUMBER
M55302/142B



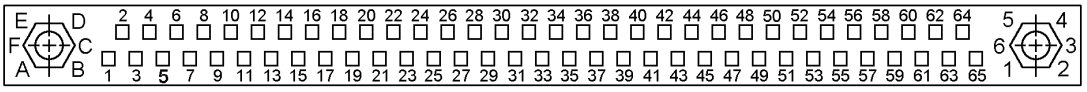
PART NUMBER
M55302/142C



PART NUMBER
M55302/142D



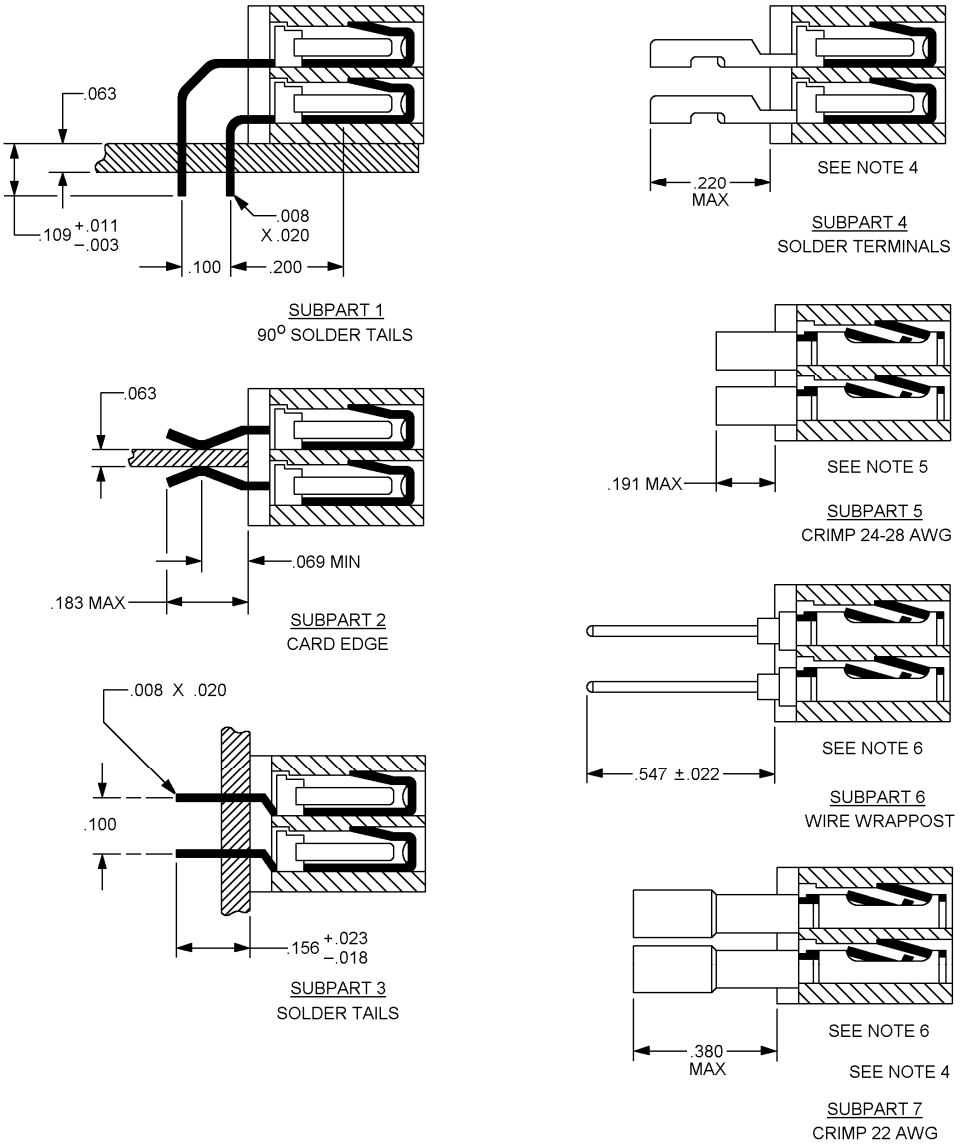
PART NUMBER
M55302/142E



PART NUMBER
M55302/142F

FIGURE 3. Contact arrangements and keying arrangements.

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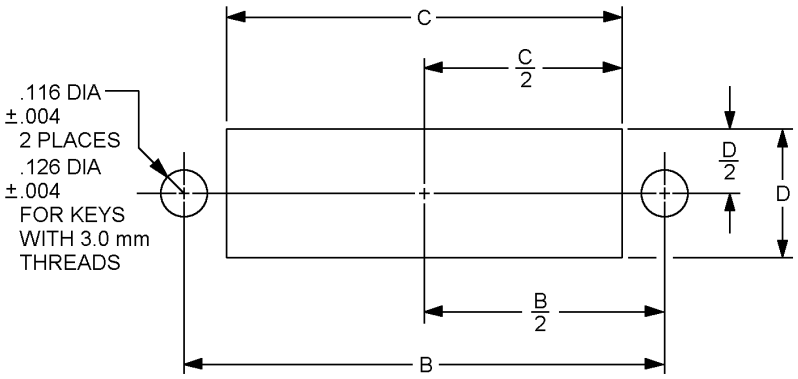


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).
4. Contact accepts no. 22 AWG wire.
5. Contact, crimp type, accepts no 22-26 AWG wires.
6. Contact with .025 inch^2 (0.64 mm^2) wrappost.

Inches	mm	Inches	mm
.003	0.08	.100	2.54
.008	0.20	.109	2.77
.011	0.28	.156	3.96
.018	0.46	.183	4.65
.020	0.51	.191	4.85
.022	0.56	.200	5.08
.023	0.58	.220	5.59
.063	1.60	.380	9.65
.069	1.75	.547	13.89

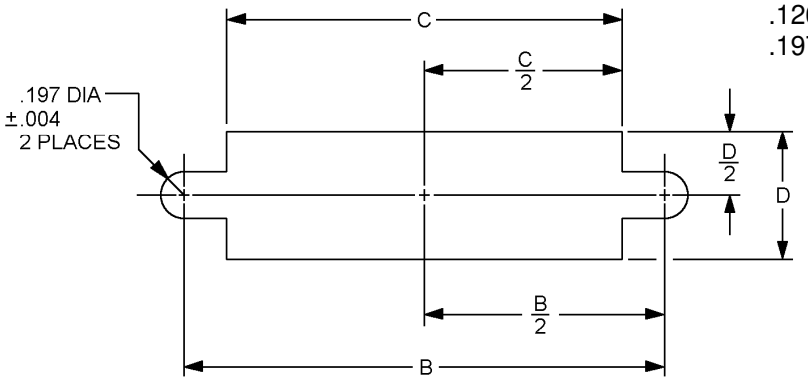
FIGURE 4. Connector terminations.

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FIXED MOUNT

Inches	mm
.004	0.10
.116	2.95
.126	3.20
.197	5.00



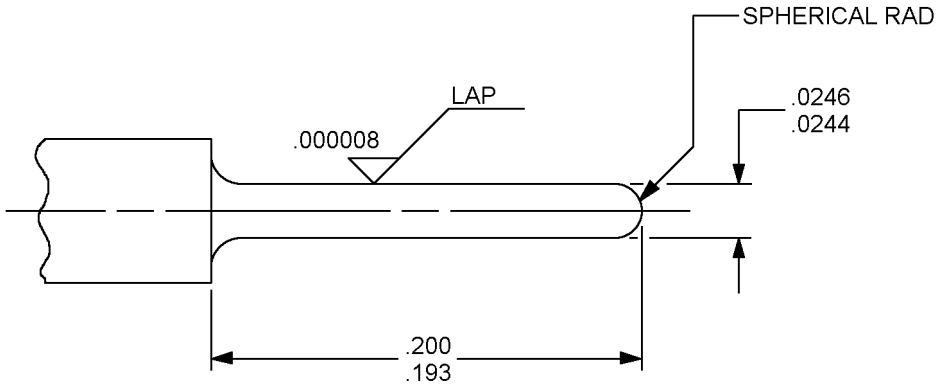
FLOATING MOUNT

Number of Contacts	17	29	33	41	53	65
B ± .008 (± 0.20)	1.200 (30.48)	1.800 (45.72)	2.000 (50.80)	2.400 (60.96)	3.000 (76.20)	3.600 (91.44)
C minimum	1.020 (25.91)	1.618 (41.10)	1.819 (46.20)	2.221 (56.41)	2.819 (71.60)	3.422 (86.92)
D minimum	.374 (9.45)	.374 (9.45)	.374 (9.45)	.374 (9.45)	.374 (9.45)	.374 (9.45)

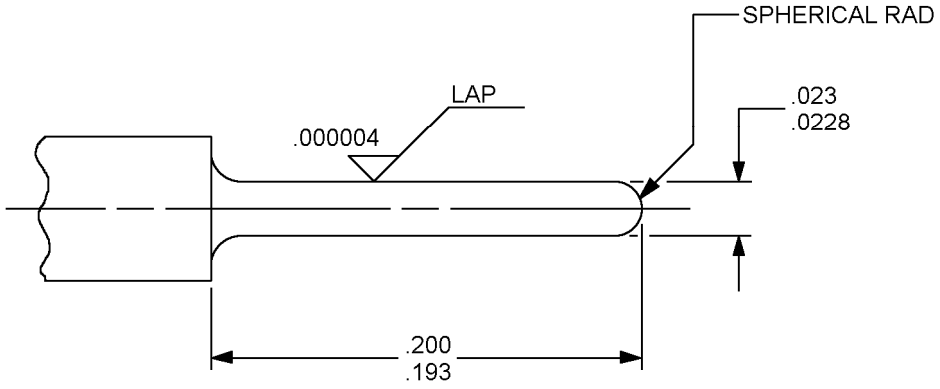
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ± .005 (0.13 mm).
4. Floating mount panel thickness .085 (2.15 mm) maximum.

FIGURE 5. Recommended panel cutouts.

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TEST PIN A



TEST PIN B

Material: Tool steel

Inches	mm
.000004	0.00001
.000008	0.00002
.0228	0.579
.023	0.58
.0244	0.619
.0266	0.676
.193	4.90
.200	5.08

FIGURE 6. Test pins.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 through 5.

Contact identification: See figure 3.

Contact retention: 4 pounds minimum for each contact.

Contact engagement: 0.7 ounce minimum, 3.6 ounces maximum.

Contact separation: 0.5 ounce minimum. Use the following test procedure:

a. Insert and withdraw test pin “A” (see figure 6)

b. Insert test pin “B” (see figure 6) fully, then withdraw the pin and measure the separation force.

Contact resistance: Shall not exceed 12 milliohms.

Dielectric withstanding voltage:

Sea level: 1,000 Vrms, 60 Hz.

High altitude: 300 Vrms, 60 Hz.

Current rating, maximum: 3.0 amperes.

Keying socket: Two hexagonal keyed guides provide 36 polarizing positions. Keys are constructed in accordance with M55302/154. Connectors are supplied with hex pins in the “1” and “A” positions, except for 1/4-turn locking styles, which are keyed in “F” and “6” positions. When required, connectors can be easily keyed to the desired position by using a standard wrench or screwdriver.

Crimping tools (for subparts 4 and 6):

Basic tool: M22520/2-01.

Positioner: M22520/2-36.

Hood types: See MIL-DTL-55302/155.

Mounting hardware: See MIL-DTL-55302/154.

Mating connectors: Shall conform to MIL-DTL-55302/140.

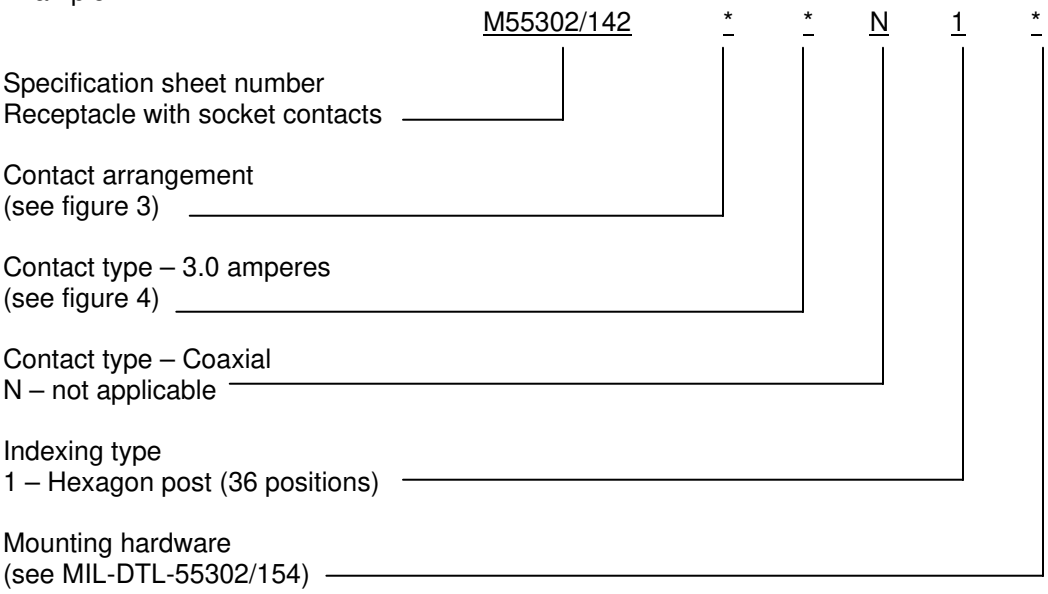
Retention clip: Optional.

Oversize pin exclusion test: Not applicable.

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Part or identifying number (PIN): Consists of the letter M, the basic number of the specification sheet, and a dash number compiled as follows:

Example:



Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/140
MIL-DTL-55302/154
MIL-DTL-55302/155

CONCLUDING MATERIAL

Review activities:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-4487-000)

Review activities:
Army – AR, AT, AV, CR4
Navy – AS, OS, MC, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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